

Procedure Overpayment: The Complete Category Framework

Inpatient DRG Payment Integrity | Category taxonomy, guideline map, validation flowchart, concept derivation methodology, and the swap-pattern data revisit

Every procedure-based DRG overpayment is a failure of one of eight validation questions. The eight questions form a cascade — an auditor (or a selection model) walks a billed ICD-10-PCS code through them in order, and the first question that fails defines the finding category, the guideline citation, and the correction (drop or swap). This framework is exhaustive by construction: a PCS code has exactly seven characters plus its grouper attributes, so any coding error must live in substantiation, multiplicity, one of the five clinically-meaningful characters, or the DRG mechanics layer. The sixteen concepts already delivered, and the twelve additional concepts in the companion workbook, are all instances of these eight categories.

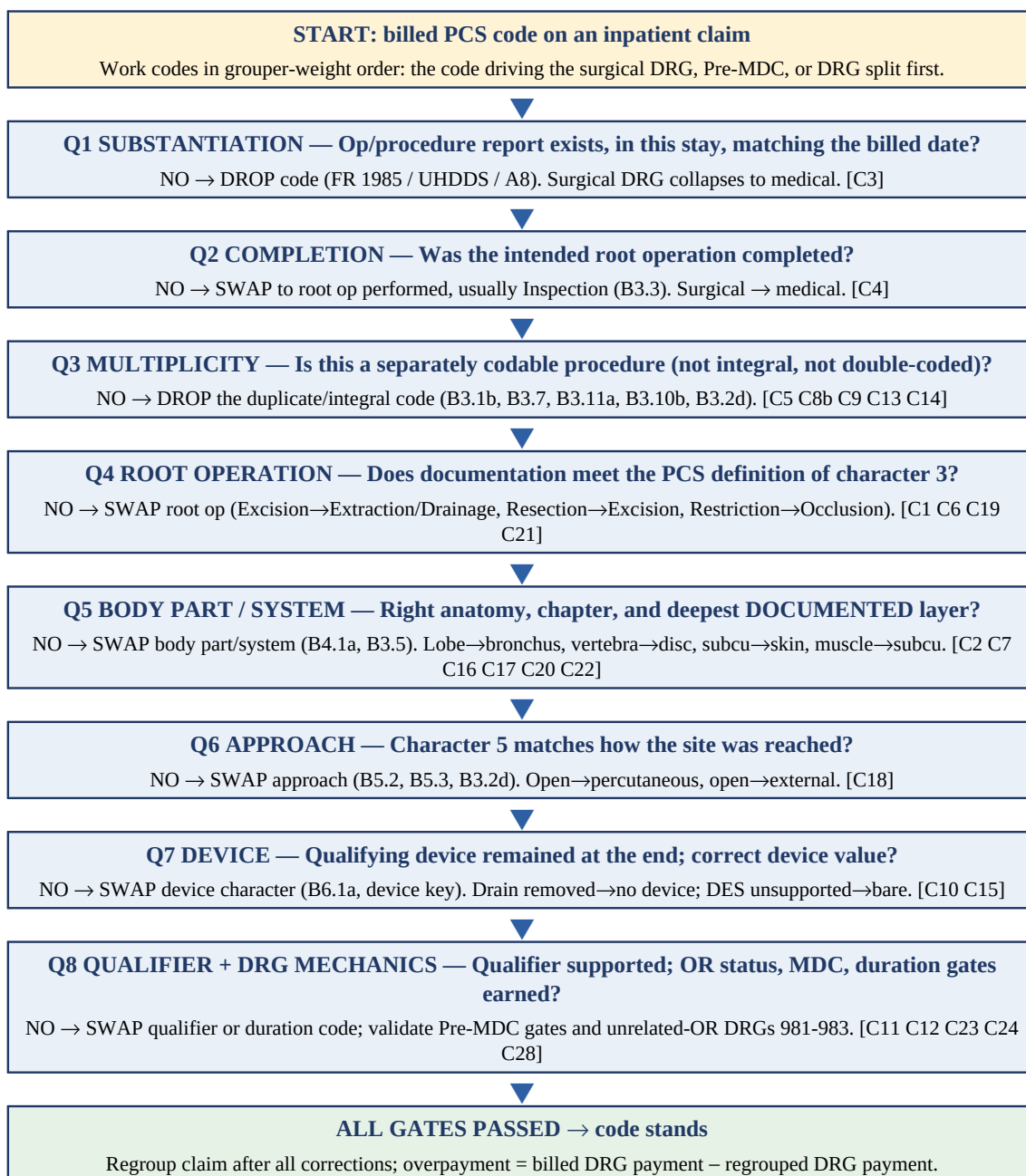
1. The Eight Categories and Their Guideline Anchors

#	Category (validation question)	Guideline anchors	Typical findings / example concepts
Q1	SUBSTANTIATION — did the procedure happen, during this stay, as billed?	Federal Register 31-Jul-1985; UHDDS 1984 (record must substantiate DRG); PCS A8	No operative report; op report from another encounter; date mismatch. Concepts C3, part of C11b.
Q2	COMPLETION — was the intended operation finished?	PCS B3.3 (discontinued procedures coded to root op performed, else Inspection)	Aborted PCI/ERCP billed as full Dilation/Repair. Concept C4.
Q3	MULTIPLICITY — coded once, and only the codable components?	PCS B3.1b (integral steps not coded); B3.2 (multiple procedures); B3.7 (Control); B3.11a (Inspection); B3.10b (fusion device hierarchy); B4.3 (bilateral values)	Control + definitive; Inspection + definitive; lap-converted double-code; fusion graft stacking; liner exchange as Removal+Replacement. Concepts C5, C8b, C9, C13, C14.
Q4	ROOT OPERATION (character 3) — does the documented action meet the PCS definition?	PCS root operation definitions; A11 (no assumptions); B3.8 (Excision vs Resection)	Excisional vs non-excisional debridement; Resection vs Excision; Excision vs Drainage (BAL); Restriction vs Occlusion. Concepts C1, C6, C19*, C21*.
Q5	BODY PART / BODY SYSTEM (characters 2+4) — right anatomy, right layer, right chapter?	PCS B4.1a (specific site); B3.5 (deepest documented layer); B4.2; body-part keys	Muscle/bone depth unsupported; lung lobe vs bronchus; vertebra vs disc; skin vs subcutaneous; nerve vs spinal cord; intracranial vs carotid. Concepts C2, C7, C16, C17*, C20*, C22*.
Q6	APPROACH (character 5) — how the operative site was actually reached?	PCS B5.2 (external); B5.3 (percutaneous endoscopic); B3.2d (converted approach)	Open billed for percutaneous carotid extirpation (thrombectomy billed as endarterectomy); external repairs coded open. Concepts C5, C18*, part of C17*.
Q7	DEVICE (character 6) — did a qualifying device remain at the end?	PCS B6.1a (device remains after procedure); device key; B3.10b hierarchy	Drainage device without retained drain; drug-eluting vs bare stent; synthetic substitute vs liner. Concepts C10, C15, part of C9.
Q8	QUALIFIER + DRG MECHANICS (character 7 + grouper) — right qualifier, right OR status, right MDC, right duration gate?	PCS qualifier definitions; Coding Clinic 4Q-2014 (vent hours); MS-DRG Definitions Manual (OR status, MDC logic, Pre-MDC gates, unrelated-OR DRGs 981-983)	Diagnostic X vs therapeutic Z; detachment level; bifurcation; short-term heart-assist qualifier; vent >96h; trach Pre-MDC gate; unrelated OR procedure DRGs. Concepts C11, C12, C23*, C24*, C28*.

** concepts marked with an asterisk are the new additions delivered in the companion workbook.*

2. The Validation Flowchart

Walk every billed PCS code (highest-weight first) through the cascade. The first NO is the finding; regroup and value. A YES at every gate means the code stands.



3. How the Sixteen Concepts Were Derived (Methodology)

Step 1 — Profile the findings universe. Of 27,337 audit rows, 3,590 are ICD-10-PCS procedure findings: 3,360 deletions with no replacement, 213 PCS-to-PCS swaps, and the rest unclear/mixed. Deletions were clustered by the first three characters (section + body system + root operation) and swaps were analyzed character-by-character.

Step 2 — Cluster deletions by prefix and comment language. The dominant deletion clusters: 0JB subcutaneous excisional debridement (~950), 0JH device insertion (256), 0KB muscle debridement (147), 0QS/0PS fracture reposition (~170), 0JD extraction (120), 0Y6 amputation (97), 0BB lung excision (87), 0SR joint replacement (72), plus vent-duration codes. Coder comments were mined for the recurring rationale: no operative report, depth not documented, procedure not performed as described, wrong body part.

Step 3 — Decompose the 213 swaps by changed character position. Each swap pair was diffed character-by-character (section 2 of the data revisit, next page). This exposed which PCS characters actually fail in production coding — body part and body system dominate, followed by approach and root operation.

Step 4 — Map every cluster to its guideline anchor. Each empirical cluster was matched to the PCS Official Guideline convention that adjudicates it (B3.3, B3.5, B3.7, B3.8, B3.10, B3.11, B4.1a, B5.x, B6.1a), to UHDDS/Federal Register substantiation, or to MS-DRG grouper mechanics. A cluster with no defensible anchor was discarded — an audit concept that cannot cite its rule does not survive appeal.

Step 5 — Generalize and extend. Each anchored cluster was generalized from the specific codes observed to the full code family the rule governs (e.g., the observed 0BBG8ZX→0BB88ZX swap became the all-lobes, all-stations bronchus concept C7). Then the guideline set itself was walked to add mechanisms the rules make inevitable but the sample had not yet surfaced (fusion device hierarchy, Control unbundling, PCI artery count).

Step 6 — Rank by expected value and split selection from audit. Concepts were ordered by DRG differential × expected hit frequency, and every concept was given two SQL layers: a claims-only selection query (dx-procedure mismatch, code-pair structure, LOS fragility) and an MR-stage audit query (note text, query compliance).

4. Data Revisit — What the 213 PCS Swap Pairs Actually Show

You asked for a revisit of the swap patterns. The 7,930 total swap rows are mostly diagnosis swaps; exactly 213 are PCS-to-PCS procedure swaps. Diffing each pair character-by-character gives the true error anatomy of procedure coding in this book of business:

PCS character (position)	Involved in N of 213 swaps	What it means	Concepts it feeds
Body part (4)	111 (52%)	The single most fragile character. Lobe vs bronchus, vertebra vs disc, specific artery, wrong laterality.	C7, C16, C20*, C22*
Body system (2)	87 (41%)	Chapter-crossing errors: subcutaneous (0J) vs skin (0H), bones (0P/0Q) vs joints (0R/0S), nerves (01) vs cord (00).	C16, C17*, C22*
Approach (5)	82 (38%)	Open vs percutaneous vs external. Includes the vent-duration family and carotid extirpation direction errors.	C11, C18*
Root operation (3)	57 (27%)	Definition failures: Excision vs Drainage vs Extraction, Resection vs Excision, Restriction vs Occlusion.	C1, C6, C19*, C21*
Qualifier (7)	32 (15%)	Diagnostic X vs therapeutic Z; short-term heart-assist qualifier; detachment level.	C12, C23*, C24*
Device (6)	25 (12%)	Synthetic substitute vs liner; stent type; drainage device.	C9, C10, C15

Named swap clusters found in the revisit (with counts) that were NOT yet concepts:

Subcutaneous vs skin repair (26 pairs): 0JQ/0JB open-approach codes swapped to 0HQ/0HB external — ED laceration repairs and lesion excisions coded one chapter too deep and one approach too invasive. This is the largest unexploited cluster and becomes Concept C17.

Vertebra vs disc (20 pairs): confirms Concept C16 at scale — biopsies and drainage on 0P/0Q swapped to 0R/0S.

Carotid extirpation approach (11 pairs): 03C codes flipping between percutaneous (3) and open (0) — thrombectomy vs endarterectomy. Open extirpation is the CEA-family surgical claim. Becomes Concept C18.

Intracranial vs carotid artery dilation (5 pairs): 037G (intracranial) swapped to 037K/L (internal carotid) — stroke-intervention body-part errors with craniotomy-family DRG exposure. Becomes Concept C20.

Lymphatic root-op ladder (5 pairs): 07T Resection → 07B Excision → 07D Extraction on lymph node sampling. Becomes Concept C21.

Lung excision vs drainage / BAL (4 pairs): 0BB diagnostic excision swapped to 0B9 drainage — bronchoalveolar lavage billed as biopsy. Extends C7 into Concept C19.

Nerve vs spinal cord release (4 pairs): 01N peripheral-nerve release swapped to 00N spinal-cord release and vice versa — decompression chapter errors. Becomes Concept C22.

Short-term heart-assist qualifier (4 pairs): 02HA3RZ swapped to 02HA3RJ — Impella-type devices missing the short-term qualifier, with heart-assist DRG-family exposure. Becomes Concept C23.

Diagnostic vs therapeutic qualifier (subset of 32): X↔Z flips on drainage and excision. Becomes Concept C24.

The twelve new concepts (C17–C28) built from these clusters plus two guideline-driven additions — bilateral body-part values (B4.3) and the unrelated-OR-procedure DRG 981–983 guard — are specified with full logic, guideline anchors, DRG impact, and selection SQL intents in the companion workbook (Additional_Procedure_Concepts.xlsx).